OFFICIAL COLLEGE PROJECT ABSTRACT · PROJECT 24 OF 55

Early Detection of Financial Fraudulent Transactions Using Graph Neural Networks

Domain: Graph Neural Networks & Cybersecurity | **Module:** project_24_financial_fraud_gnn

OFFICIAL ABSTRACT

The rapid growth of digital payments and online banking has increased the number of financial transactions performed every day, while also creating new opportunities for fraudulent activities. Traditional fraud detection systems often analyze transactions individually using predefined rules or conventional machine learning techniques. However, financial fraud is frequently connected to relationships between multiple accounts, devices, locations, and transactions, making it difficult to identify suspicious behavior by examining a single transaction alone. The proposed Early Detection of Financial Fraudulent Transactions Using Graph Neural Networks (GNNs) aims to address this challenge by analyzing the relationships and behavioral patterns within financial transaction networks. The system is developed using Python, machine learning, and Graph Neural Network techniques. Financial entities such as customers, bank accounts, payment methods, devices, and transactions can be represented as nodes and relationships between them as edges. This graph-based representation enables the system to study not only individual transaction characteristics but also the connections between different entities. The model can learn patterns associated with legitimate and suspicious transaction behavior and assign a fraud probability to new transactions. The major uniqueness of the proposed system is its relationship-aware fraud detection approach. Instead of treating every transaction as an independent event, the system examines connected transaction patterns to identify unusual behavior. For example, multiple accounts interacting with the same suspicious account or unusual transaction connections between different entities can provide additional evidence of potential fraud. This helps the model detect complex fraud patterns that may remain hidden when only individual transaction details are analyzed. Another important feature is early fraud prediction. The system aims to identify suspicious patterns at an earlier stage, allowing potentially fraudulent transactions to be flagged before significant financial damage occurs. A risk score can be generated for each transaction so that high-risk activities can be prioritized for further verification

or investigation. The proposed system can be applied in digital banking, online payment platforms, e-commerce, and financial service applications. By combining graph-based relationship analysis with deep learning, the project provides a scalable approach for detecting evolving fraud patterns. Overall, the project demonstrates how Graph Neural Networks can transform financial fraud detection from transaction-level analysis into relationship-level intelligence, enabling earlier identification of suspicious activities and supporting more effective financial security systems.

📌 Project Summary & System Specifications

Project Title	Early Detection of Financial Fraudulent Transactions Using Graph Neural Networks
Specialization / Category	Graph Neural Networks & Cybersecurity
Target Technology Stack	Python (Flask), PyTorch / TensorFlow / Scikit-Learn, Modern Web UI
Default Backend Port	`:5024`
Local Implementation	`project_24_financial_fraud_gnn`